

Project Thesis — Contribution Summary

Embedded Control and User Interface Development

Applicant: I-En Lee

Institution: Friedrich-Alexander-Universität Erlangen-Nürnberg (FAU)

Project Thesis: *User Interface and Control Software for a Automated Test Bench for the Analysis of Biofilm Formation*

Project objective

Develop embedded control software and a touchscreen interface for an automated test bench supporting long-term investigations of biofilm formation on urological implants, with programmable micturition cycles and environmental monitoring.

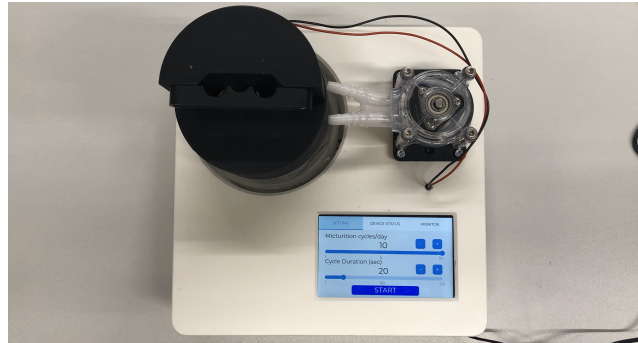


Figure 1. Integrated automated test bench. The embedded control software and touchscreen interface developed in this project enable experiment configuration, cyclic actuation, and operational monitoring.

My contributions

- Developed modular embedded C/C++ software on the Arduino GIGA R1 WiFi, integrating actuator control, sensor acquisition, and user-interface functions.
- Implemented peristaltic-pump and UV-LED control, with configurable daily cycles and pump-active duration, using `millis()`-based cycle scheduling.
- Developed an LVGL touchscreen interface for experiment settings, device status, cycle counting, and temperature visualization, including confirmation before stopping an experiment.
- Integrated BMP280 temperature acquisition and automatic I²C bus recovery to restore sensor communication following transient interruptions.

Evaluation documented in the Project Thesis

At the time of thesis writing, the platform had operated continuously for nearly two months and completed more than 400 automated test cycles without critical software errors or control failures. The interface was assessed through an expert-based heuristic evaluation using Nielsen's ten usability principles, without external participants.

Contribution to the accompanying conference manuscript

The embedded control software and touchscreen interface developed in this project enabled programmable cyclic operation and standalone monitoring of the test bench, supporting the long-term experiments reported in the accompanying co-authored manuscript:

An Automated In-Vitro Platform for Long-Term Evaluation of Endourological Medical Devices — Real-Time Anomaly Detection.